Awni Altabaa

PhD Candidate · Statistics & Data Science, Yale University

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I work on the foundations of artificial intelligence, developing new architectures, algorithms, and theoretical frameworks to identify simple, unifying principles that explain how intelligent systems learn.

RESEARCH

Architectural Inductive Biases
What mechanisms an architecture needs for models to learn efficiently and form robust abstractions.
[3] [5] [7] [8] [9] [10]

Reasoning in Transformers & LLMs
How models learn, represent, and execute multi-step reasoning, and how to improve out-of-distribution generalization.
[1] [2] [3] [4]

Theoretical Foundations
Formal accounts of how architecture, learning algorithm, and problem structure determine what models can learn.
[1] [4] [6] [7] [11]

EDUCATION

Ph.D. in Statistics & Data Science, Yale University, New Haven, CT — 2022 – 2027 (expected)
Advisor: John Lafferty

Master's in Statistics, Yale University, New Haven, CT — 2022 – 2024

Bachelor's in Applied Mathematics, Queen's University, Kingston, ON — 2018 – 2022

EXPERIENCE

Research Intern, Microsoft Research, AI Interaction & Learning group — Summer 2026
Paper: *Putting Transformer Theories to the Test* [1]

Quantitative Researcher Intern, Citadel Securities — Summer 2025
Systematic Equities / FICC. End-to-end time-series forecasting alpha research: data collection, feature generation with an LLM agent system (RAG), and alpha modeling.

PUBLICATIONS

1. **Putting Transformer Theories to the Test**
   A. Altabaa, T. Schnabel, and K. Tomlinson
   arXiv preprint, 2026 arXiv:2610.32104

2. **Uncovering Latent Reasoning Strategies in Language Models**
   A. Altabaa and J. Lafferty
   arXiv preprint, 2026 arXiv:2607.17674

3. **Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning**
   A. Altabaa, S. Chen, J. Lafferty, and Z. Yang
   arXiv preprint, 2025 Blog arXiv:2510.14095

4. **CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision**
   A. Altabaa, O. Montasser, and J. Lafferty
   Neural Information Processing Systems (NeurIPS), 2025 · Spotlight Blog arXiv:2505.15927

5. **Disentangling and Integrating Relational and Sensory Information in Transformer Architectures**
   A. Altabaa and J. Lafferty
   International Conference on Machine Learning (ICML), 2025 Blog arXiv:2405.16727

6. **On the Role of Information Structure in Reinforcement Learning for Partially-Observable Sequential Teams and Games**
   A. Altabaa and Z. Yang
   Neural Information Processing Systems (NeurIPS), 2024 arXiv:2403.00993

7. **Approximation of Relation Functions and Attention Mechanisms**
   A. Altabaa and J. Lafferty
   Information Theory, Probability and Statistical Learning: A Festschrift in Honor of Andrew Barron, 2024
   arXiv:2402.08856

8 **Learning Hierarchical Relational Representations through Relational Convolutions**
A. Altabaa and J. Lafferty
Transactions on Machine Learning Research (TMLR), 2024 Blog arXiv:2310.03240

9 **The Relational Bottleneck as an Inductive Bias for Efficient Abstraction**
T. W. Webb, S. M. Frankland, **A. Altabaa**, S. Segert, K. Krishnamurthy, D. Campbell, J. Russin, T. Giallanza, *et al.*
Trends in Cognitive Sciences (TICS), 2024 arXiv:2309.06629

10 **Abstractors and Relational Cross-Attention: An Inductive Bias for Explicit Relational Reasoning in Transformers**
A. Altabaa, T. W. Webb, J. D. Cohen, and J. Lafferty
International Conference on Learning Representations (ICLR), 2024 Blog arXiv:2304.00195

11 **Decentralized Multi-Agent Reinforcement Learning for Continuous-Space Stochastic Games**
A. Altabaa, B. Yongacoglu, and S. Yüksel
IEEE American Control Conference (ACC), 2023 arXiv:2303.13539

HONORS

Rising Stars in Statistics & Data Science, Cornell University — Sep 2026

Trainee, NSF AI Institute for Artificial and Natural Intelligence (ARNI) — 2023 – present
Funded 2023 – 2025 (advisor: John Lafferty)

Annie Bentley Lillie Prize in Mathematics, Queen's University — 2022

Nellie & Ralph Jeffery Award in Mathematics, Queen's University — 2021

Wm. Roy Hardick Scholarship, Queen's University — 2021

TALKS & PRESENTATIONS

Rising Stars in Statistics & Data Science, Cornell University — Poster · Sep 2026

Formal Languages and Neural Networks (FLaNN) Seminar — Invited talk · Feb 2026
Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning [3]

NeurIPS 2025, San Diego — Spotlight · Dec 2025
CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision [4]

Formal Languages and Neural Networks (FLaNN) Seminar — Invited talk · Oct 2025
CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision [4]

Deep Learning: Classics and Trends (DLCT), ML Collective — Invited talk · May 2025
Disentangling and Integrating Relational and Sensory Information in Transformer Architectures [5]

Two Sigma PhD Symposium — Invited talk · Jun 2024

New England Statistics Symposium (NESS 2024) — Contributed talk · May 2024
On the Role of Information Structure in Reinforcement Learning for Partially-Observable Sequential Teams and Games [6]

Yale Wu Tsai Institute Workshop on Abstraction and Relational Reasoning — Contributed talk · Nov 2023
Learning Hierarchical Relational Representations through Relational Convolutions [8]

American Control Conference (ACC 2023), San Diego — Contributed talk · Jun 2023
Decentralized Multi-Agent Reinforcement Learning for Continuous-Space Stochastic Games [11]

TEACHING EXPERIENCE

Teaching Fellow, Yale University — 2023 – 2026
Introductory Machine Learning (S&DS 2650/5650) — Spring 2026
Intermediate Machine Learning (S&DS 3650/6650) — Fall 2025
Theory of Reinforcement Learning (S&DS 685) — Spring 2025
Introductory Machine Learning (S&DS 265/565) — Fall 2023

Teaching Assistant, Queen's University — 2020 – 2022
Linear Algebra, Differential Calculus, Integral Calculus

SERVICE

Reviewer: NeurIPS (2024, 2025, 2026), ICLR (2025, 2026), ICML (2025, 2026; Gold Reviewer Award, 2026), Journal of Data Science